

Exam Date & Time: 06-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Bigdata Analytics [IT-4034]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the key features of the Hadoop Distributed File System (HDFS) and its role in the Hadoop ecosystem. Provide a brief description of each feature and its significance in distributed storage and processing. Additionally, discuss how HDFS contributes to fault tolerance and scalability in big data applications. (5)
 - A)
 - B) Illustrate the significance of each Big Data characteristic in addressing possible challenges posed by the "MIT_IT student" dataset. (3)
 - C) Describe the process of reading data to HDFS from a client's perspective, detailing the specific steps involved and the interactions between the client and the HDFS architecture. (2)
- 2) Illustrate the essential phases of a typical word-count MapReduce job. Describe the arguments taken in the classes for mappers and reducers. Write a sample code for a reducer class. (5)
 - A)
 - B) "Businesses use predictive analytics models to forecast trends and consumer behaviour based on social media data", Justify. Provide examples of predictive analytics applications in marketing or product development. (3)
 - C) Explore how Spark's resilient distributed datasets (RDDs) and directed acyclic graph (DAG) execution engine enable fault tolerance, data parallelism, and lazy evaluation in Spark applications. (2)
- 3) Illustrate the concept of RDD with suitable examples. Consider an RDD containing a collection of natural numbers. Describe how you would use transformations and an action to identify and count the even numbers. (5)
 - A)
 - B) What are the different types of NoSQL databases? Read the following code and identify the type of NoSQL database. Also arrange the data in a tabular form.

DB.STUD.INSERT(
{
NAME : "ROHIL",
AWARDS: [{ AWARD : "DANCING", RANK: "1ST", YEAR: 2008 },
{ AWARD : "DRAWING", RANK: "3RD", YEAR: 2010 } ,
(3)

```

{AWARD : "SINGING", RANK: "1ST", YEAR: 2015 } ],
RNO: 9,
ADDRESS: BLR
}
NAME : "REYA",
AWARDS: [ { AWARD : "DANCING", RANK: "1ST", YEAR: 2008 },
{AWARD : "DRAWING", RANK: "3RD", YEAR: 2010 } ,
{AWARD : "SINGING", RANK: "1ST", YEAR: 2015 } ],
ADDRESS: HYD
}
)

```

- C) Illustrate the architecture of Spark SQL. (2)
- 4) You are facilitating a workshop for a team of data engineers and business analysts at an international data summit. Your session aims to compare and contrast Apache Hive and Pig, two pivotal tools in big data analytics. Discuss at least four key features of each tool and explain the different operational modes available in Hive and execution modes in Pig. Additionally, using suitable diagrams, illustrate the data flow in Hive and the architecture of Pig. How do these tools complement each other in handling large-scale data projects, and what are their unique strengths? (5)
- A)
- B) You are hosting a workshop for junior developers at a software development firm to enhance their understanding of basic learning algorithms. Discuss the foundational concepts and applications of various learning algorithms, including Supervised, Unsupervised, and Reinforcement Learning. Provide examples to illustrate how each algorithm type is used to tackle specific problems in software development and data analysis. What are the key strengths and limitations of each algorithm, and how do they impact practical implementation in projects? (3)
- C) Illustrate the fault tolerance mechanisms employed by Apache Spark to ensure reliable communication and data consistency in distributed environments. (2)
- 5) Illustrate the concept of Binomial Logistic regression. With a sample code illustrate the concept of Local Vector data type in MLlib. (5)
- A)
- B) NoSQL: What is it? Provide scenarios to demonstrate why NoSQL is necessary. (3)
- C) What are the major components of Spark Streaming? Illustrate with a flow diagram. (2)

-----End-----